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Machine Learning — Assignment Answers

Q1. Blocks of a Machine Learning System, with Supervised and Unsupervised Learning Examples (5 Marks)

General Architecture of a Machine Learning System

A typical machine learning system is built from a sequence of functional blocks that together take raw data and convert it into a usable predictive model. The diagram below shows the standard pipeline.

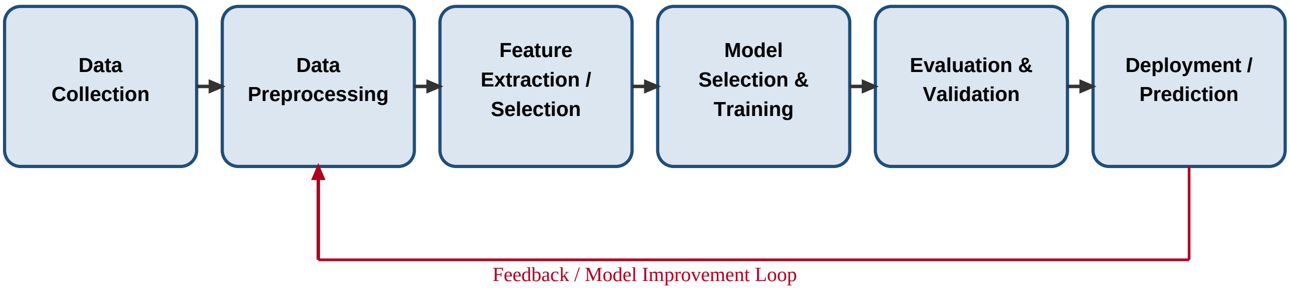

Fig. 1 — General block diagram of a Machine Learning system

Explanation of each block:

- **Data Collection:** Gathering raw data from sensors, databases, files, or the web that is relevant to the problem.
- **Data Preprocessing:** Cleaning the data — handling missing values, removing noise/outliers, normalizing or scaling values, and encoding categorical data.
- **Feature Extraction / Selection:** Identifying and selecting the most relevant attributes (features) that influence the outcome, and/or transforming data into a suitable feature representation.
- **Model Selection & Training:** Choosing a suitable algorithm (e.g., Decision Tree, SVM, Neural Network, K-Means) and fitting it to the training data so that it learns the underlying pattern.
- **Evaluation & Validation:** Testing the trained model on unseen data using metrics such as accuracy, precision, recall, F1-score, or RMSE, and validating using techniques like k-fold cross-validation.
- **Deployment / Prediction:** Using the finalized model to make predictions on new real-world data, and feeding results back for continuous improvement (feedback loop).

Example — Supervised Learning

In supervised learning, the model is trained on labeled data, i.e., every input example is paired with a known correct output. For example, classifying an email as Spam or Not Spam using a dataset where each past email is already tagged with its correct class.

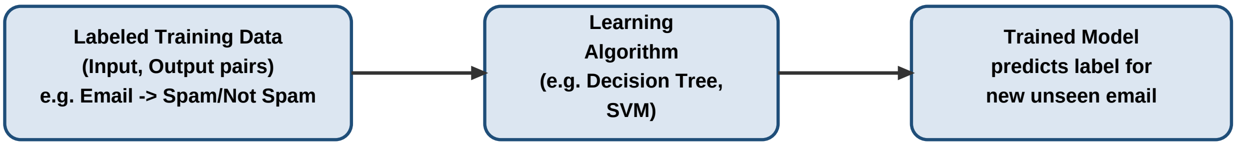

Fig. 2 — Supervised learning scheme (email spam classification)

Example — Unsupervised Learning

In unsupervised learning, the model is given unlabeled data and must discover hidden structure or groupings on its own. For example, grouping customers into segments based on purchasing behaviour without any predefined labels, using an algorithm such as K-Means clustering.

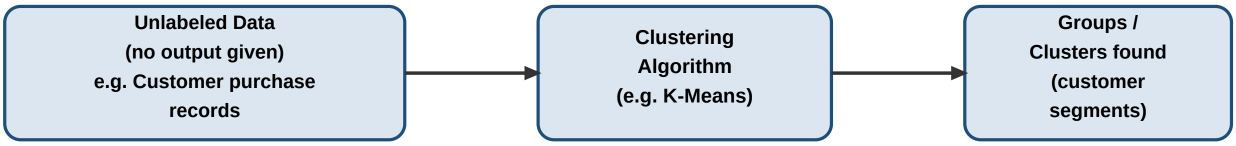

Fig. 3 — Unsupervised learning scheme (customer segmentation)

Q2. Machine Learning System for Disaster Management (Flood / Earthquake Prediction) (5 Marks)

The goal is to design a system that ingests historical records together with real-time sensor data to predict the likelihood of disaster events such as floods and earthquakes, and to trigger timely alerts.

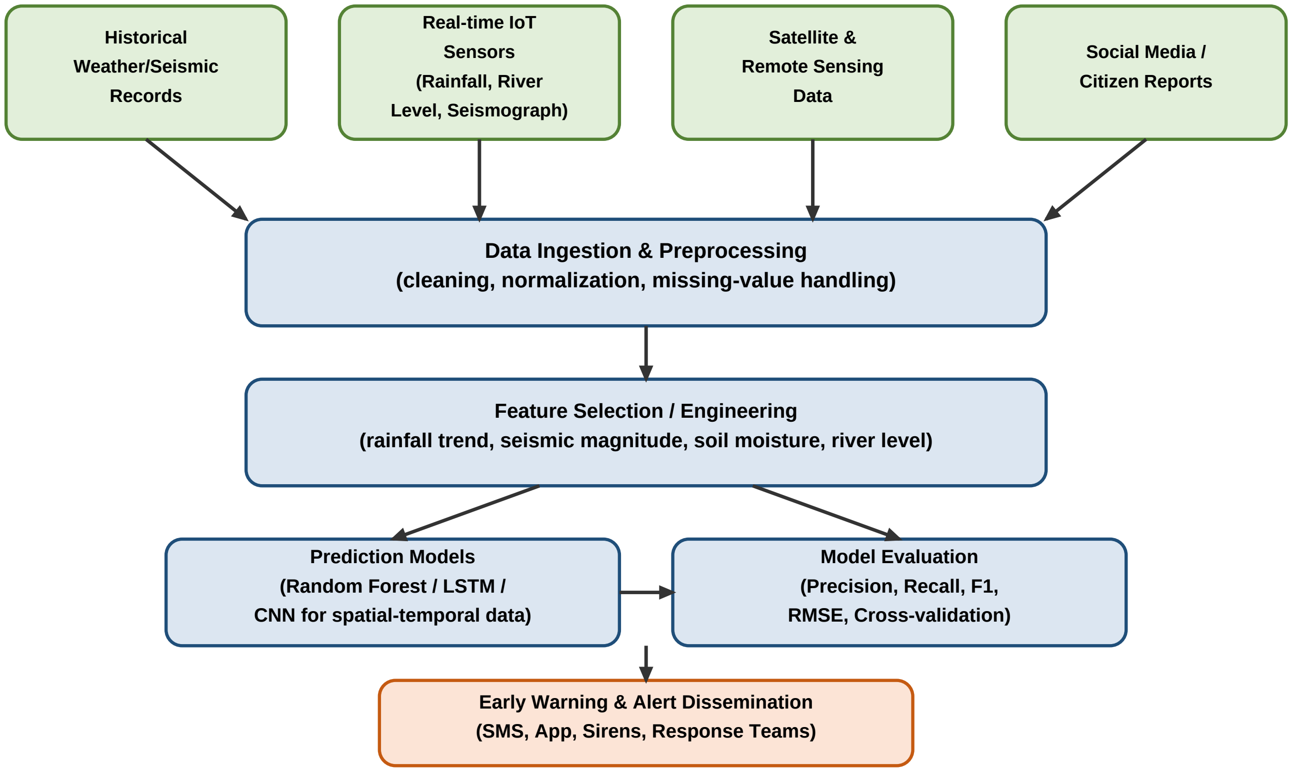

Fig. 4 — System architecture for disaster prediction and early warning

Choice of Model

Because disaster-related data (rainfall, river level, seismic tremors) is largely time-series in nature, a combination of models is preferred:

- **Random Forest / Gradient Boosting (e.g., XGBoost):** Robust for tabular historical data with mixed features; gives feature-importance for interpretability, useful for flood-risk classification.
- **LSTM (Long Short-Term Memory) networks:** Well suited to sequential, real-time sensor streams (rainfall trend, river-level trend) where the temporal pattern matters, useful for flood forecasting.
- **CNN or hybrid CNN-LSTM:** Useful when spatial data such as satellite imagery or seismic waveform data is involved, e.g., earthquake precursor pattern detection.

Data Preprocessing

- Handling missing sensor readings using interpolation or forward-fill.
- Removing noise and outliers from sensor data (e.g., faulty seismograph spikes).
- Normalizing/scaling numerical features (rainfall in mm, magnitude in Richter scale) to a common range.
- Synchronizing timestamps across multiple heterogeneous data sources (satellite, IoT, historical archives).
- Encoding categorical/geographic information such as region or soil type.

Feature Selection

- Correlation analysis to identify which variables (rainfall intensity, river discharge, soil saturation, tectonic plate movement, past seismic frequency) are most predictive of an event.
- Dimensionality reduction (e.g., PCA) when a large number of sensor channels are available.
- Domain-driven features: cumulative rainfall over 24/48/72 hours for floods; seismic gap and historical fault activity for earthquakes.

(a) Training, Validation and Evaluation Steps

- **Split the data:** Divide historical data into training, validation, and test sets (e.g., 70/15/15), preserving time-order for time-series data.
- **Train the model:** Fit the chosen model (e.g., LSTM/Random Forest) on the training set, tuning hyperparameters such as learning rate, number of trees, or sequence length.
- **Validate:** Use the validation set (or k-fold / walk-forward cross-validation for time-series) to tune hyperparameters and avoid overfitting.
- **Test/Evaluate:** Assess final performance on the held-out test set using metrics such as Accuracy, Precision, Recall, F1-score (for event/no-event classification) and RMSE/MAE (for continuous predictions like water level).
- **Iterate:** Retrain periodically as new real-time data arrives, using the feedback loop shown in Fig. 1.

(b) Role in Early Warning and Emergency Response

- The trained model continuously scores incoming real-time sensor data; when the predicted risk crosses a defined threshold, an alert is automatically generated.
- Alerts are disseminated through SMS, mobile apps, sirens, and dashboards to citizens and disaster-response authorities.
- Predicted severity and location help emergency teams prioritize evacuation routes, allocate resources (rescue teams, medical aid), and pre-position relief supplies before the event peaks.
- The system supports post-event analysis by comparing predictions with actual outcomes, which feeds back into retraining for improved future accuracy.

Q3. Find-S Algorithm — Play Tennis Dataset (5 Marks)

The Find-S algorithm starts with the most specific hypothesis h = <Ø,Ø,Ø,Ø> and generalizes it only using the **positive** examples, ignoring all negative examples.

Positive examples in the dataset: Example 3, 4, 5, 7

Step	Example Used	Outlook	Temperature	Humidity	Wind	Hypothesis after this step
Initial	-	-	-	-	-	<Ø, Ø, Ø, Ø>
1	Ex 3 (Overcast, Hot, High, Weak, Yes)	Overcast	Hot	High	Weak	<Overcast, Hot, High, Weak>
2	Ex 4 (Rainy, Mild, High, Weak, Yes)	differs→?	differs→?	High=High	Weak=Weak	<?, ?, High, Weak>
3	Ex 5 (Rainy, Cool, Normal, Weak, Yes)	? (same)	? (same)	differs→?	Weak=Weak	<?, ?, ?, Weak>
4	Ex 7 (Overcast, Mild, Normal, Strong, Yes)	? (same)	? (same)	? (same)	differs→?	<?, ?, ?, ?>

Final Maximally Specific Hypothesis: h = <?, ?, ?, ?>

This means that, based on the positive examples in this dataset, no single attribute value consistently determines 'Play Tennis = Yes' — the positive examples vary across every attribute, so Find-S generalizes all four attributes to 'don't care' (?).

Q4. Candidate Elimination Algorithm — Enjoy Sport Dataset (5 Marks)

The Candidate Elimination algorithm maintains two boundaries of the version space: the specific boundary S (generalized using positive examples) and the general boundary G (specialized using negative examples).

Initialization:

S0 = <Ø, Ø, Ø, Ø>   G0 = <?, ?, ?, ?>

Example	Type	Update to S	Update to G
Warm, Normal, Weak → Yes	Positive	S1 = <Sunny, Warm, Normal, Weak>	G1 = <?, ?, ?, ?> (unchanged)
Warm, High, Strong → Yes	Positive	Humidity & Wind differ → generalize: S2 = <Sunny, Warm, ?, ?>	G2 = <?, ?, ?, ?> (unchanged)
, Cold, High, Weak → No	Negative	S3 = S2 = <Sunny, Warm, ?, ?> (unchanged)	Specialize G to exclude example, consistent wi G3 = { <Sunny,?,?,?>, <?,Warm,?,?> }
Cold, Normal, Weak → Yes	Positive	Temperature differs → generalize: S4 = <Sunny, ?, ?, ?>	Remove members inconsistent with exampl <?,Warm,?,?> removed (Cold≠Warm) G4 = { <Sunny,?,?,?> }
Warm, Normal, Strong → No	Negative	S5 = S4 = <Sunny, ?, ?, ?> (unchanged)	<Sunny,?,?,?> already excludes this example (Rainy≠Sunn G5 = { <Sunny,?,?,?> }

Final Version Space: S = { <Sunny, ?, ?, ?> }   and   G = { <Sunny, ?, ?, ?> }

Since the specific boundary S and the general boundary G have converged to the same hypothesis, the version space contains exactly one hypothesis: h = <Sunny, ?, ?, ?>, meaning 'Enjoy Sport = Yes' whenever the weather is Sunny, regardless of Temperature, Humidity, or Wind.

Q5. Find-S Algorithm — Buys Product Dataset (5 Marks)

Applying Find-S using only the positive examples (Example 3 and Example 4) and ignoring the negative examples (Example 1 and Example 2).

Step	Example Used	Age	Income	Student	Credit Rating	Hypothesis after this step
Initial	-	-	-	-	-	<Ø, Ø, Ø, Ø>
1	Ex 3 (Middle, High, No, Fair, Yes)	Middle	High	No	Fair	<Middle, High, No, Fair>
2	Ex 4 (Senior, Medium, No, Fair, Yes)	differs→?	differs→?	No=No	Fair=Fair	<?, ?, No, Fair>

Final Maximally Specific Hypothesis: h = <?, ?, No, Fair>

This means that, based on the given positive examples, a customer is predicted to buy the product when Student = No and Credit Rating = Fair, irrespective of Age or Income.